MORPHISMS OF COMPLETED COHERENT SYSTEMS ON NOETHERIAN SPACES ARE THE COMPLETION OF THE INTERNAL HOM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9655-9667 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9669-9675, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Formal-functions suite member feeding the algebraization cards (L16158 same wave, L16140 shelf).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale descent for the coherent formal category on spaces; Cohomology of Schemes, lemma completion-internal-hom (the schemes identification).

Source reference. Original lemma 0882: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9675–9675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S and let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. Let $\mathcal{F}$, $\mathcal{G}$ be coherent $\mathcal{O}_X$-modules. Set $\mathcal{H} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$. Then*

$$\lim H^0(X, \mathcal{H}/\mathcal{I}^n\mathcal{H}) = \mathrm{Mor}_{Coh(X,\mathcal{I})}(\mathcal{F}^\wedge, \mathcal{G}^\wedge).$$

Proof. Since $\mathcal{H}$ is a sheaf on $X_{étale}$ and since we have étale descent for objects of $Coh(X,\mathcal{I})$ it suffices to prove this étale locally. Thus we reduce to the case of schemes which is Cohomology of Schemes, Lemma 0882. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.